

Calendar-ProductType recommendation app (Jonathan SGP <> AiGENThix) – Phase 2

Required Files for project

Core Application & Logic

- **.env** – Environment variables
- **app.py** – Main application entry point
- **data_cleaner.py** – Data preprocessing logic
- **rules.py** – Business rules and logic
- **llm_parser.py** – Logic for parsing LLM responses
- **client_plan_llm.py** – LLM-driven client planning logic
- **agent_controller.py** – Orchestration of various agents
- **llm_router.py** – Routing requests to appropriate LLM completion logic

Tools & Modules

- **calendar_tools.py** – Calendar integration utilities
- **client_tools.py** – Client-specific data operations
- **data_manager.py** – Management of data flow within the app
- **email_sender.py** – Email notification module
- **whatsapp_sender.py** – WhatsApp Business API integration
- **scheduling_tools.py** – Task and appointment scheduling logic
- **preview_engine.py** – Engines for previewing outputs

Data & Documentation

- **JonathanProj.ipynb** – Main project notebook
- **data.xlsx** – Primary input data
- **cleaned_data.xlsx** – Preprocessed output data
- **reminders.xlsx** – Stored reminder data
- **recommendationOutput.xlsx** – Generated recommendations
- **jonathan-writing-profile.txt** – Writing style configuration
- **marketoutlook-temporary.txt** – Temporary market research data
- **writing-style-prompt.txt** – Base prompt for writing consistency

Configuration & Dependencies

- **requirements.txt** – Python package dependencies
- **.env_example** – Template for environment variables

Project Dependencies

- VS Code
- VS Code Jupyter extensions
- Python 3.10 or any version after 3.10
- All libraries inside requirements.txt

Setup Operations (One time only):

1. How to Setup Virtual Environment

- How to open the project folder in VS code

- Open VS code
- Click on **File** option in the top bar



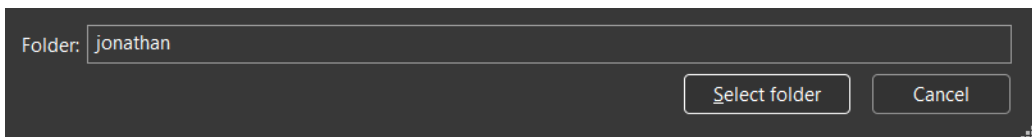
3. Select **Open Folder**



- Go to the folder in which you have setup the files



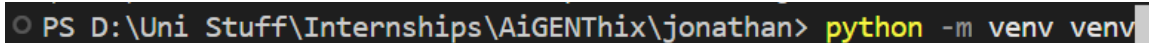
- Click on Select folder



- How to Setup Virtual Environment

- To open VS code terminal, use the shortcut **CTRL + J**
- In the terminal, type the following command to initialize a python environment

```
python -m venv venv
```

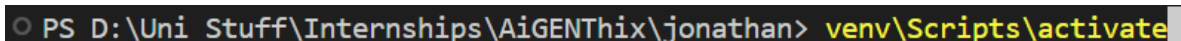


2. How to Activate Virtual Environment

- NOTE: You do not always have to create a virtual environment as long as a folder named 'venv' exists in your project folder. But, everytime you run the code, you need to always activate the virtual environment.**

- In the VS code Terminal, input the following command

```
venv/Scripts/activate
```

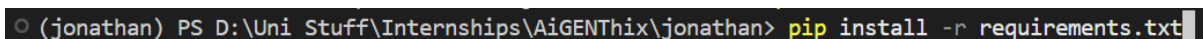


3. How to install dependencies

- NOTE: You do not always have to install dependencies, as long as you have done it once in your venv its fine.**

- Activate Virtual Environment (venv/Scripts/activate)**
- In the terminal, use the following command to install required libraries (might take some time)

```
pip install -r requirements.txt
```



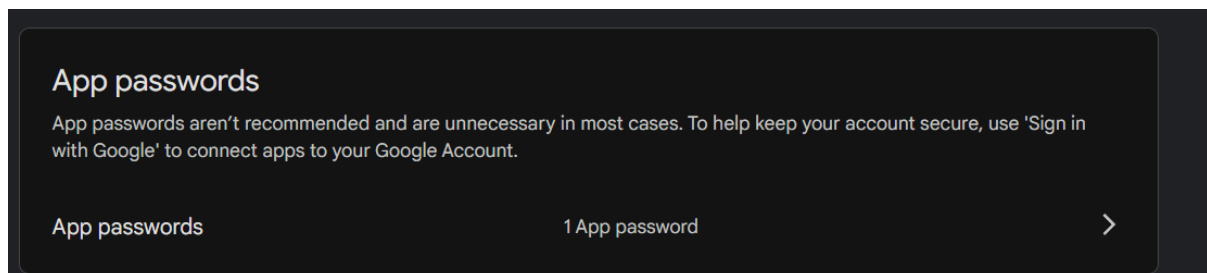
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4. Setup .env file

1. Open the .env file in VS code
2. It should look something like this:

```
1  GEMINI_API_KEY=ENTER KEY
2  GEMINI_MODEL=gemini-2.5-flash
3
4  GMAIL_EMAIL=ENTER EMAIL
5  GMAIL_APP_PASSWORD="ENTER APP PASSWORD"
6
7  WHATSAPP_API_TOKEN=your_meta_access_token_here
8  WHATSAPP_PHONE_NUMBER_ID=your_phone_number_id_here
```

3. In line 1, after the '=', paste your Gemini API key
4. In line 4, enter your Email address from which you want the mails to be sent.
5. In line 5, you must enter your Gmail app password, this is different from the actual password for your gmail account. This can only be gotten after activating 2 Step verification on your gmail Id.
 - a. First activate 2 step verification on your gmail account
 - b. Then, go to app passwords page:

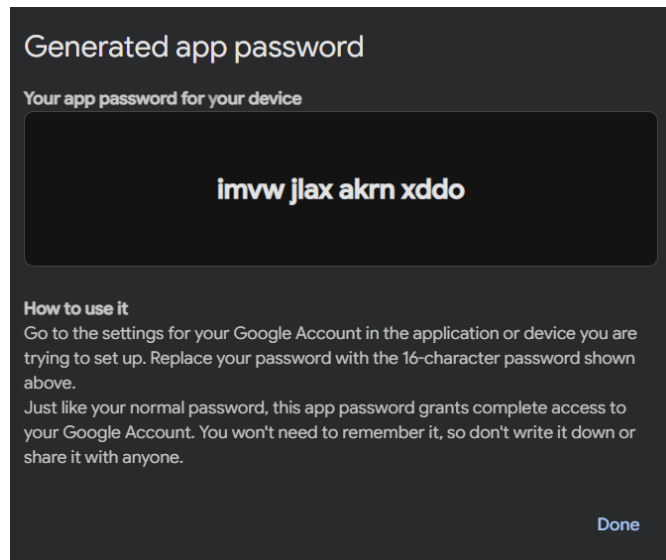


- c. Add a new app password name

A screenshot of the 'Create new app-specific password' form. The form has a dark background. At the top, it says 'To create a new app-specific password, type a name for it below...'. Below this is a text input field labeled 'App name' with the text 'JonathanProj' entered. At the bottom right, there is a 'Create' button.

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- d. Click on create, it will show you an OTP, copy that, and **paste the OTP in line 5** after “=”



6. In lines 7 and 8, enter your whatsapp Business API credentials
7. Save the file (ctrl + S)

- **Your Setup is completed, Now, you don't need to do anything except step (2) every time you want to run the code**
- **The Indicator for an Active Virtual Environment is (venv) in your command terminal**

```
(venv) PS D:\Uni Stuff\Internships\AiGENThix\TestJonathanInstructions>
```

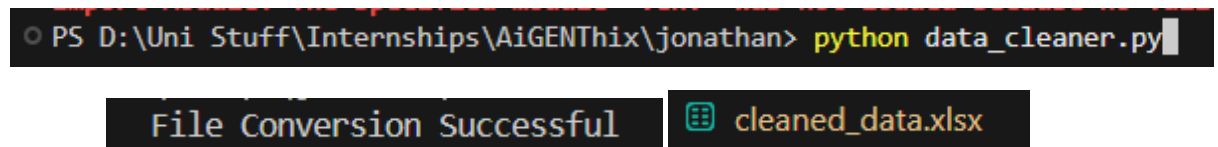
Running the code and application

1. How to clean data

1. First, make sure that **"data.xlsx"** is the name of the file in which you have the raw data
2. Paste that file inside the project folder
3. **Activate Virtual Environment if you haven't (venv/Scripts/activate)**
4. In the terminal, type the following command

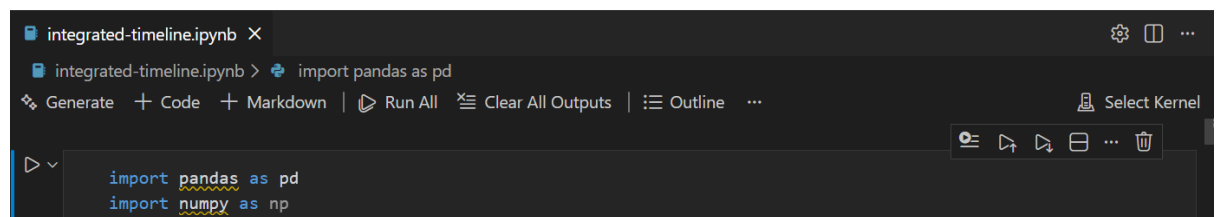
```
python data_cleaner.py
```

5. This will output a new file called: 'cleaned_data.xlsx'

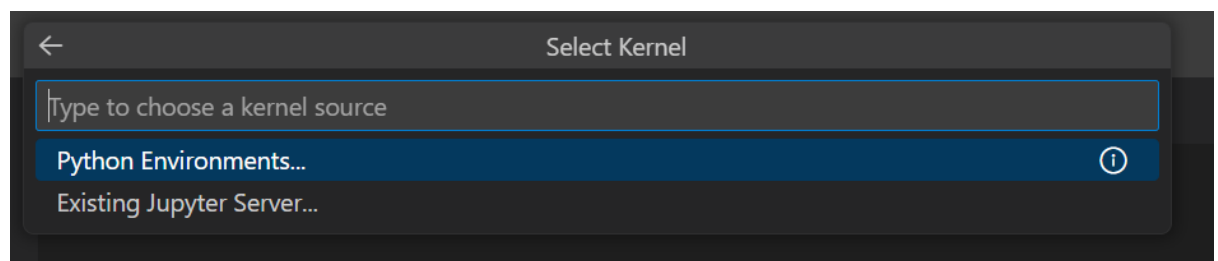


2. How to run analytics and Machine Learning Models

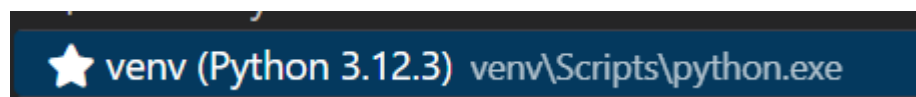
1. First, make sure that **"cleaned_data.xlsx"** is present in your project folder
2. Open the file **'JonathanProj.ipynb'**
3. This is a Jupyter notebook, now, select your **venv** as the kernel.
4. First click on 'Select Kernel'. If it already shows **'venv'** then you may skip step 4, 5 and 6.



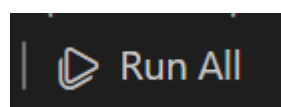
5. Click on 'Python Environments'



6. Click on 'venv'



7. Click on the 'Run all' button



8. This will output a new file called: **'recommendationOuput.xlsx'**

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3. How to run Calendar application (Streamlit application)

1. First activate Virtual Environment if you haven't (venv/Scripts/activate)
2. In the terminal, type the following command

```
streamlit run app.py
```

3. If you didn't directly get redirected, then go to your preferred browser and search for:

```
http://localhost:8501
```
